ReturnLens

An evidence-checked returns agent for small direct-to-consumer sellers.

USER

A seller who manages orders and returns alone.

SCOPE IN 24 HOURS

Return triage only, mock store data.

WHAT IT IS NOT

Another order-status chatbot.

CORE PIPELINE: Video → Plan → Evidence → Verify → Approve

Team Rookie · 3 members · PS-04 submission

PROBLEM CONTEXT

The return decision is made from memory, before the evidence is checked.

— **Today the seller opens many tabs:** watch the video, check the order, check shipping, remember the SKU and the return window, then decide alone.

— **A wrong decision costs both ways:** a wrong refund loses money and stock, an unfair denial loses a repeat customer.

— **Sourced signal:** Fraud estimates differ by definition: 9% (NRF 2025, fraud only) vs 15.14% (Appriss/Deloitte 2024, fraud + abuse).

— **The gap is not a missing returns portal:** It is that no single record ties order, video frame, policy clause, and decision together.

END-TO-END WORKFLOW

Verify first. Act only after approval.

1. Intake

Buyer video, 30-60 seconds, mock web chat

Ex: video + order_id + reason

→

2. Plan

Typed plan, schema-validated

Ex: typed JSON, requires_seller_approval: true

→

3. Evidence

Mock order & shipping APIs, plus timestamped video frames

Ex: order, shipping, frame 00:07

→

4. Verify

Deterministic policy in code, plus an independent critic

Ex: P-01 to P-08, critic veto only

→

5. Decide

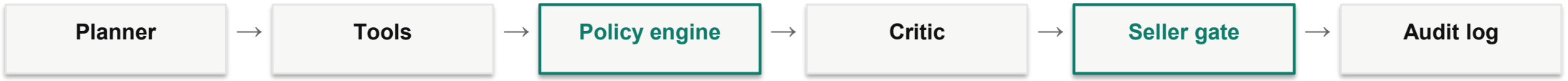
Seller approves in one click, or the case escalates

Ex: label + draft pending, or escalate

Returns only in 24 hours. Plus one tiny order-status route for PS-04 fit: ask order, call order and shipping APIs, return a grounded status.

SYSTEM ARCHITECTURE

Policy checks and seller approval control actions



- **Tools:** Mock store API, mock shipping API, video indexer.
- **Policy clauses:** Delivered, within 7 days, exact SKU match, tag if required, defect visible with a frame if claimed, video quality gates, refund cap, no contradiction.
- **Model calls:** Exactly 3 base calls per case (planner, vision, critic) plus retries.
- **Frames:** Server-generated frame list with frame_id, timestamp, video hash, and frame hash. The model may only cite frames it was given.
- **Critic:** The critic reviews cited frames and flags unsupported observations.
- **Fail-closed rule:** Weak evidence escalates. Never auto-deny. Never auto-refund.
- **Injection limit (honest):** Injection cannot change the law, exceed the server cap, or execute an action alone, but it can change an observation field, so the seller must check the frames in the evidence pack.

DECISION STRIP: Contradiction → Escalate | Hard fail → Escalate | Unclear only → Ask more | All pass → Approve for seller review

VERIFICATION & DEMO EVIDENCE

Every decision can be clicked back to a frame, a clause, and a log entry.

CASE B (AUDIT PROOF: ESCALATED)

Order: HEAD-WHT-02, White

Frame 00:07: HEAD-BLK-09, Black

Policy: P-03 SKU mismatch, FAIL

Result: **ESCALATE**, no label, no refund draft, no denial

Seller options: Accept as mis-shipment / Ask for outside-box label / Hold for inspection.

- **Case A, approved for review:** ORD-1042 (₹1,499), SKU matches, tag present, within window. Seller approval creates a mock return label and a refund draft in pending_return_receipt, not an executed refund.
- **Case C, ask for more evidence:** 4-second dark video, brightness 28, only 2 usable frames. Case stays open with an exact re-film checklist.
- **Case D, why video beats photos:** One continuous take from opening a sealed parcel to showing the label in one continuous sequence.

Staged sample videos, mock data, no real customer data. If the model fails or the quota runs out, escalate. Do not approve without evidence.

DELIVERY & GOVERNANCE

Buildable in 24 hours by 3 people, if the scope stays narrow.

24-HOUR PLAN & SCOPE CUTS

- **Person A:** Backend, agent graph, policy, critic, audit log.
- **Person B:** Video pipeline, video indexer, seller console, evidence pack view.
- **Person C:** Integration, mock data, 4 demo cases, fallback testing, demo video, README.
- **Project logic:** Written inside the 24 hours. Before that, only research, design, accounts, and library choice.
- **Cut first:** WhatsApp, exchanges, multi-seller, audio, public deployment.
- **Never cut:** Policy engine, critic, seller gate, audit log, 4 demo cases.

RISKS AND GUARDS

- **Video unclear:** Ask for more evidence or escalate.
- **Quota or timeout:** Cached analysis with the correct cache key, or escalate.
- **Duplicate or stale approval:** One return process per order in MVP, label and refund draft both idempotent in one transaction, approval bound to case version, evidence hash, policy version, action, and amount.
- **Refund cap (server-computed):** 0 < amount <= min(₹4,999, remaining refundable amount, order paid amount)

PS-04 checklist: mock tool calls, refund-limit guardrails, action log, human confirmation before refund.

No idea can guarantee top-100 selection. This deck claims a checkable, buildable agent, not a guaranteed result.